

Robust Cooperative LiDAR Perception with Uncertainty-Aware Fusion and Spatial TrackFormer Association

Progress Report

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Abstract

This report describes a cooperative 3D perception pipeline for OPV2V under localization and communication noise. A PointPillars detector provides local vehicle proposals, BELT-style evidential and covariance-aware fusion recovers substantial accuracy under noise, and a spatial adaptation of TrackFormer produces cross-agent object-association embeddings. On the official noisy OPV2V test set, uncertainty-aware BELT fusion improves AP@0.7 from 0.294 for corrected naive late fusion to 0.415. Adding the current TrackFormer association improves strict localization further to 0.423, while slightly reducing AP@0.5; this identifies association recall as the main remaining bottleneck.

1 Problem and Motivation

Cooperative perception lets several connected autonomous vehicles (CAVs) share detections so that an ego vehicle can see beyond its own occlusions. Late fusion is attractive because agents transmit compact object-level messages rather than dense feature maps. Its weakness is that detections from different agents must be expressed in a common coordinate frame, associated correctly, and fused despite localization error and asynchronous observations.

We study PointPillars late fusion on the OPV2V benchmark [1]. The noise setting follows the BELT-Fusion repository: 0.2m Gaussian position noise, 0.2° heading noise, and 100ms communication delay. The research questions are: (i) can uncertainty-aware fusion recover detection accuracy under this noise; and (ii) can learned cross-agent association improve over geometry-only association?

2 PointPillars and the BEV Representation

Each CAV observes a raw LiDAR point cloud $\mathcal{P}_i = \{(x, y, z, r)\}$ in its local coordinate system. PointPillars partitions the ground plane into fixed XY cells; all points in the same cell, over the full vertical range, form one *pillar* [2]. The pillar feature network applies a learned point-wise encoding and pooling operation to produce one learned descriptor for every nonempty pillar. These descriptors are scattered back onto an XY grid and processed by a 2D convolutional BEV backbone.

For our PointPillars configuration, the backbone output is

$$F_i \in \mathbb{R}^{384 \times 100 \times 176}. \quad (1)$$

The 100×176 dimensions are the backbone’s spatial BEV grid after its convolutional downsampling/upsampling design; they are not the raw LiDAR point count. The 384 channels are learned

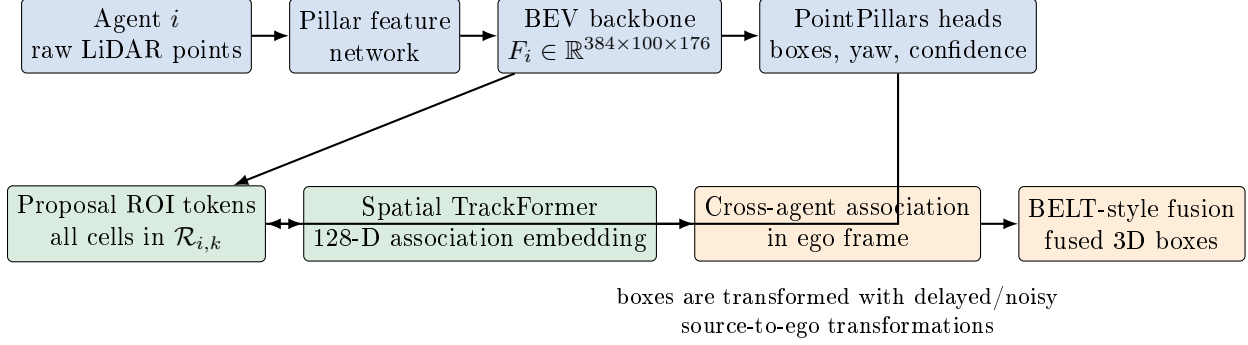


Figure 1: Implemented cooperative-perception pipeline. Each agent extracts local BEV features and detections. Spatial TrackFormer converts all vehicle-covered BEV cells into an association embedding, while BELT-style fusion uses uncertainty to combine associated boxes.

feature channels, formed by concatenating three 128-channel backbone branches. Detection heads applied to F_i predict vehicle boxes, orientations, confidence scores, and dense anchor outputs.

For a predicted box $b_{i,k}$, let $\mathcal{R}_{i,k}$ be the set of BEV backbone cells covered by its rotated footprint. The first embedding baseline used only one 384-D cell selected at the proposal anchor. The improved cache retains *all* cells $\{F_i(r) : r \in \mathcal{R}_{i,k}\}$, typically 6–21 cells, plus their relative locations, the box, confidence, matched ground-truth identity, and the agent transformation matrix. Importantly, BEV features remain local to each agent; it is the predicted box geometry that is transformed to the ego frame before cross-agent geometric comparison.

3 Coordinate Alignment and Noise

Let $T_{e \leftarrow i} \in SE(3)$ transform a source agent’s local coordinates into the ego frame. For a local proposal box, the center and yaw are aligned before association as

$$\tilde{b}_{i,k}^{(e)} = T_{e \leftarrow i}^{\text{noisy, delayed}} b_{i,k}. \quad (2)$$

The delayed/noisy transform is essential: a prior OpenCOOD late-fusion path retained source boxes but used a clean transformation matrix during fusion, thereby bypassing the intended pose perturbation. We corrected this path so non-ego boxes are placed using the delayed, noisy pose.

This correction produces the valid robustness baseline: clean late fusion has $\text{AP@0.7} \approx 0.78$, whereas noisy naive late fusion has $\text{AP@0.7} = 0.294$. The drop is expected: small pose errors are amplified at high IoU because several agents’ boxes no longer align precisely.

4 BELT-Style Uncertainty-Aware Late Fusion

The public BELT-Fusion snapshot documents the method but does not provide a complete runnable OPV2V/PointPillars path or released trained weights. We therefore implemented an OpenCOOD-compatible adapter that retains the published design principles [4]: evidential classification uncertainty, heteroscedastic box uncertainty, uncertainty-aware confidence fusion, and covariance-weighted box fusion.

For classification, an evidential head predicts nonnegative evidence e_c and Dirichlet parameters $\alpha_c = e_c + 1$. The total evidence $S = \sum_c \alpha_c$ determines an uncertainty proxy $u \propto 1/S$: low

evidence yields high uncertainty. For regression, the head predicts a diagonal covariance $\Sigma_{i,k}$ for box parameters. Associated box estimates are combined by precision weighting,

$$\Sigma_f^{-1} = \sum_i \Sigma_i^{-1}, \quad \mu_f = \Sigma_f \sum_i \Sigma_i^{-1} \mu_i, \quad (3)$$

so an uncertain observation has less influence on the fused box. The implementation additionally accounts for known source pose-noise variance in the ego frame.

This works because localization noise is not equally reliable across all detections. Rather than averaging inconsistent boxes uniformly, the fusion stage downweights detections whose learned or propagated uncertainty is high.

5 From Temporal TrackFormer to Spatial Association

Original TrackFormer solves multi-object tracking as frame-to-frame set prediction [3]. A CNN extracts image features, a Transformer encoder forms image memory, and a decoder uses object queries. Object queries from frame $t - 1$ are propagated into frame t ; Hungarian matching, class/no-object supervision, ℓ_1 box loss, generalized-IoU loss, and false-positive track-query supervision train the system jointly.

Our setting replaces temporal dependence with a spatial one at a common timestamp. A source CAV plays the role of the previous frame and the ego CAV plays the role of the current frame. The model consumes PointPillars ROI tokens instead of image features, propagates source-object queries to the ego BEV memory, and asks whether source and ego proposals describe the same physical object. This is an adaptation of TrackFormer’s attention and query-propagation mechanism; it is not a claim that the original paper performed V2X association.

For proposal k , ROI tokens are projected and position encoded before the encoder/decoder. The final query $q_{i,k}$ is combined with explicit ego-frame box geometry $g_{i,k}$ and confidence $s_{i,k}$ to form a normalized association embedding,

$$z_{i,k} = \frac{h_\theta([q_{i,k}; g_{i,k}; s_{i,k}])}{|h_\theta([q_{i,k}; g_{i,k}; s_{i,k}])|_2} \in \mathbb{R}^{128}. \quad (4)$$

The association loss includes an identity contrastive term and hard-negative term,

$$\mathcal{L}_{\text{assoc}} = \lambda_{\text{emb}} \mathcal{L}_{\text{InfoNCE}} + \lambda_{\text{hard}} \mathcal{L}_{\text{hard}}, \quad (5)$$

where hard negatives are nearby vehicles with similar physical size. During full TrackFormer training, this is combined with detection/tracking losses,

$$\mathcal{L}_{\text{full}} = \mathcal{L}_{\text{assoc}} + \mathcal{L}_{\text{cls}} + \lambda_1 \mathcal{L}_{\ell_1} + \lambda_{\text{giou}} \mathcal{L}_{\text{GIoU}}. \quad (6)$$

The explicit geometry is necessary because BEV appearance alone is ambiguous: nearby vehicles often have similar LiDAR structure. The transformer supplies object and surrounding spatial context, while ego-frame geometry and confidence provide calibrated physical constraints.

6 Experimental Results

All end-to-end AP results below use the official OPV2V test split and the same noise realization. Values are shown to three decimal places.

Table 1: Noisy OPV2V late-fusion results.

Method	AP@0.3	AP@0.5	AP@0.7
Naive late fusion	0.860	0.712	0.294
BELT geometry fusion	0.851	0.752	0.415
BELT + Spatial TrackFormer	0.846	0.743	0.423

The geometry-conditioned association head reached 0.915 top-1 cross-agent association accuracy on clean validation and 0.891 on the clean official test set. Without retraining, it retained 0.907 top-1 accuracy on noisy validation. A noisy association-head fine-tuning stage achieved 0.909 on noisy validation, but did not yield a meaningful additional end-to-end test gain; the best test result remained $\text{AP@0.7} = 0.423$.

BELT provides the dominant robustness recovery: $0.294 \rightarrow 0.415$ at AP@0.7 . TrackFormer adds a smaller but held-out improvement: $0.415 \rightarrow 0.423$. The trade-off is association recall: TrackFormer forms 8.94 multi-agent groups per frame compared with 10.16 for geometry-only BELT, which explains the small AP@0.5 reduction.

7 Conclusions and Next Steps

The experiments establish a valid noisy PointPillars baseline, a working uncertainty-aware BELT implementation, and a cross-agent spatial TrackFormer association model with strong standalone identity accuracy. The remaining gap is not simply embedding quality: an embedding top-1 score does not directly optimize the precision–recall trade-off of complete fusion groups.

The next step is to evaluate association links directly against ground-truth identities on noisy validation scenes, measuring false links and missed links separately. These diagnostics can train a calibrated pairwise association score that jointly uses embedding similarity, ego-frame displacement, predicted covariance, and confidence, followed by one-to-one global assignment. This targets the observed failure mode: recover valid groups without reintroducing incorrect noisy associations.

References

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